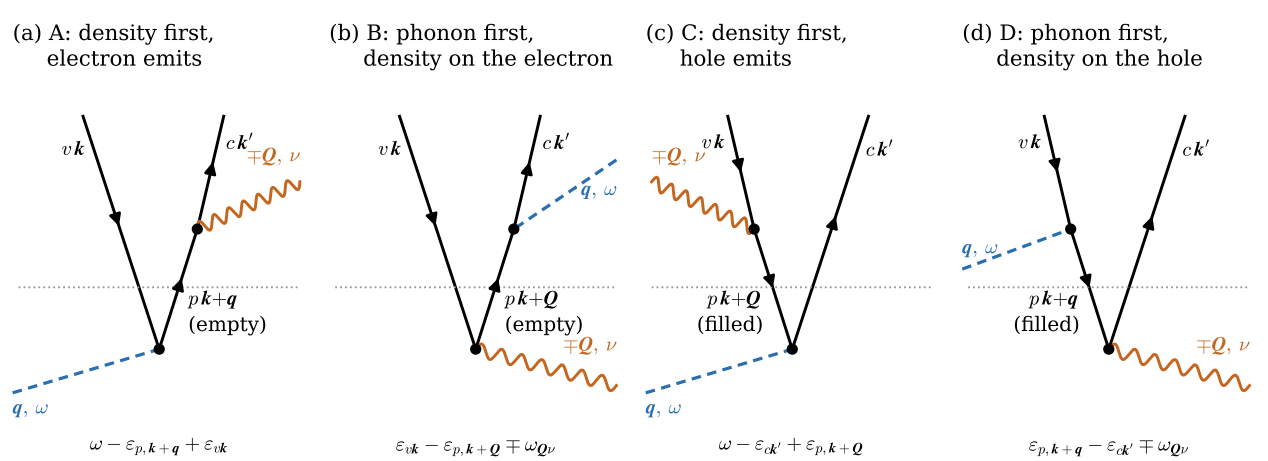
(a) A: density first,
electron emits
$v\boldsymbol{k}$
$c\boldsymbol{k}'$
$\mp\boldsymbol{Q}$, ν
$p\boldsymbol{k}+\boldsymbol{q}$
(empty)
$\boldsymbol{q}$, ω
$\omega - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} + \varepsilon_{v\boldsymbol{k}}$
(b) B: phonon first,
density on the electron
$v\boldsymbol{k}$
$c\boldsymbol{k}'$
$\boldsymbol{q}$, ω
$p\boldsymbol{k}+\boldsymbol{Q}$
(empty)
$\mp\boldsymbol{Q}$, ν
$\varepsilon_{v\boldsymbol{k}} - \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}} \mp \omega_{\boldsymbol{Q}\nu}$
(c) C: density first,
hole emits
$\mp\boldsymbol{Q}$, ν
$v\boldsymbol{k}$
$c\boldsymbol{k}'$
$p\boldsymbol{k}+\boldsymbol{Q}$
(filled)
$\boldsymbol{q}$, ω
$\omega - \varepsilon_{c\boldsymbol{k}'} + \varepsilon_{p,\boldsymbol{k}+\boldsymbol{Q}}$
(d) D: phonon first,
density on the hole
$v\boldsymbol{k}$
$c\boldsymbol{k}'$
$\boldsymbol{q}$, ω
$p\boldsymbol{k}+\boldsymbol{q}$
(filled)
$\mp\boldsymbol{Q}$, ν
$\varepsilon_{p,\boldsymbol{k}+\boldsymbol{q}} - \varepsilon_{c\boldsymbol{k}'} \mp \omega_{\boldsymbol{Q}\nu}$